BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Final GCSE Computer Science Consolidation

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What should guide the choice of a final revision topic?

2. How many colours can a bitmap with 4 bits per pixel use?
   A) 4
   B) 8
   C) 32
   D) 16

3. What happens to raw size when both bitmap dimensions double at fixed depth?

4. A microphone records digital sound. "Sampling stores every instant of the sound wave." Correct this. How does it affect recorded detail?

Today's objectives

Explain the core ideas of final gcse computer science consolidation using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Recall the full course

Paper 1 systems and Paper 2 computational thinking are equal components.

Use accurate language

State precise definitions and causal effects.

Demonstrate independence

Complete unseen traces, calculations and explanation without hints.

Check answers

Use command words, units, bounds and allocated marks before submission.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Recall the full course
Paper 1 systems and Paper 2 computational thinking are equal components.

Use accurate language
State precise definitions and causal effects.

Demonstrate independence
Complete unseen traces, calculations and explanation without hints.

Check answers
Use command words, units, bounds and allocated marks before submission.

I Do - Worked model

Solve an integrated question: a 20 x 20 bitmap with 8-bit depth is uploaded over a network. Calculate uncompressed bytes and explain one network factor.

- Pixels =20 x 20=400.
- Bits =400 x 8=3200.
- Bytes=3200 / 8=400.
- Bandwidth, latency or congestion can affect upload time; explain the chosen factor.

We Do - Guided practice

1. A list [2,6,10,14,18] is sorted. Trace binary search for 18.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Print the combined size of the first and last file in this list.

```
array filesMB = [2,6,10]
print(...)
```

You Do - Independent exam questions

Q2. Give one difference between compiler and interpreter. [3 marks]

Q3. Calculate 10101010 binary in denary. [2 marks]

Q4. Explain how to make a program robust against invalid numeric input. [4 marks]

Exit Ticket - True or False?

OCR J277 has two equally weighted examination components.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Evidence of a specific knowledge or method gap from the recent targeted review.

2. D - 16

3. The number of pixels and raw pixel-data size become four times as large.

4. Measure amplitude at regular intervals. Store each rounded (quantised) value in binary. Sample rate and bit depth limit the detail captured.

I Do - Explain the model

Pixels =20 x 20=400.

Bits =400 x 8=3200.

Bytes=3200 / 8=400.

Bandwidth, latency or congestion can affect upload time; explain the chosen factor.

We Do - Model answers

1. Found at index4.

Code First - Answer

```
array filesMB = [2,6,10]
print(filesMB[0] + filesMB[2])
```

Expected output: 12

You Do - Self-mark

Q2: Compiled source is translated to executable/intermediate output before execution of translated program; interpreted execution is mediated by an interpreter.

Q3: 170.

Q4: Use type/range checks; handle failure safely and request another value; test boundary/erroneous cases.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.